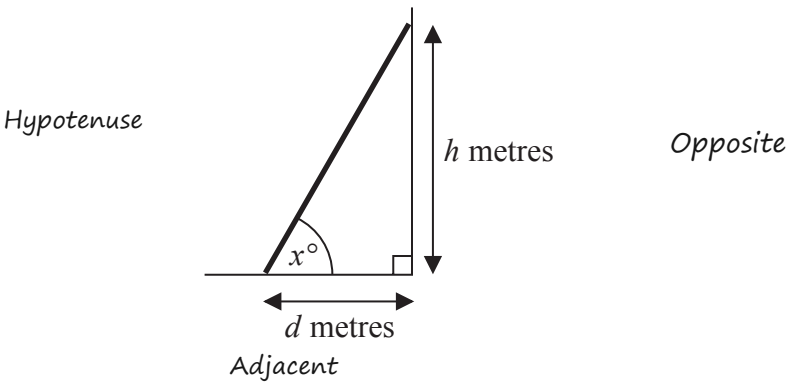
11 The bottom of a ladder is on horizontal ground.
The top of the ladder is leaning against a vertical wall.



The bottom of the ladder is d metres from the wall.
The top of the ladder is h metres above the ground.
The angle between the ladder and the ground is $x°$

Some safety instructions say it is safe to climb the ladder when

$$h = 4d$$

(a) Work out the value of x when $h = 4d$

$\tan x = \frac{h}{d} = 4$

1 mark
Opposite ÷ adjacent

$x = \tan^{-1}(4) = 75.9637...°$

1 mark
Inverse tangent

$x \approx 76.0°$

1 mark
75.9°–76°

(3)

Some different safety instructions say the angle between the ladder and the ground should be 75°
The ladder is moved so that $x = 75$

(b) How does this affect the height, h metres, of the top of the ladder above the ground?

The top is lower: a smaller angle gives a lower height.

(1)

1 mark
h decreases

(Total for Question 11 is 4 marks)